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A Modular Python Framework for Intelligent Plant Recommendation and Farming Decision Support

CSA0808 - Python Programming

Under The Supervision of:
Dr. Ponmaniraj Sambasivam

Presented By:
Soujanya. B (192521389)
Devadarshini(192521256)

ABSTRACT

This project proposes a **Python-based modular framework** to recommend suitable plants and support farming decisions using agricultural data. The system analyzes soil and environmental conditions and manages agricultural knowledge to provide accurate recommendations.

- Python-based intelligent framework
- Crop recommendation using agricultural data
- Modular architecture
- Easy to extend with Machine Learning
- Supports better farming decisions

INTRODUCTION

Choosing the right crop is essential for improving agricultural productivity. This project uses **Python** to process agricultural data and provide intelligent plant recommendations through a modular decision support system.

- Improves crop selection
- Uses soil and weather information
- Python for data processing
- Modular and scalable design
- Supports future AI integration

OBJECTIVES

The objective of this project is to develop a **Python-based modular framework** that analyzes agricultural data and provides intelligent plant recommendations for effective farming decision support.

Key Points:

•**Develop** a modular Python framework for intelligent plant recommendation.

•**Analyze** soil, weather, and crop data using Python to support decision-making.

•**Provide** accurate recommendations through efficient data processing and agricultural knowledge management.

Literature survey

S.No	Authors	Paper Title	Methods	Dataset	Accuracy	Key Findings
1	Sawant et al. (2026)	IoT-Driven Smart Crop Recommendation	GNB, IoT, ML	Soil & environmental data	99.55%	Real-time crop recommendation using IoT and ML.
2	Shastri et al. (2025)	ML & XAI for Crop Recommendation	Gradient Boosting, LIME	2,200 samples, 22 crops	99.27%	Improves accuracy and recommendation transparency.
3	Abu Bakr et al. (2025)	Learning-Based Crop Recommendation	RF, SVM, KNN, LSTM, GPT-2	2,200 samples, 22 crops	99.55%	Provides accurate and interactive farming recommendations.
4	Afzal et al. (2025)	Soil-Based ML Crop Recommendation	RF, XGBoost, Extra Trees	2,200 samples, 22 crops	98%	Soil and climate data improve crop prediction.
5	Senapaty et al. (2024)	ML-Based Crop Decision Support	13 ML classifiers, SMOTE	1,480 records, 37 crops	100%	Supports effective crop selection and farming decisions.

PROBLEM STATEMENT

Agriculture is highly dependent on factors such as **soil quality, weather conditions, and crop requirements**. Farmers often find it difficult to select the most suitable crop because this information is scattered and environmental conditions change frequently. Traditional decision-making methods may not always provide accurate or timely recommendations.

To address this challenge, this project proposes a **Python-based intelligent decision support framework** that processes agricultural data, manages crop knowledge, and generates reliable plant recommendations using a modular approach.

Key Points

•Difficulty in selecting suitable crops due to changing soil and weather conditions.

•Use Python for **data processing, analysis, and intelligent plant recommendation**.

Module 1: Plant Suitability Analysis and Decision Support Engine

Definition

This module analyzes soil and environmental parameters to determine the most suitable plant for cultivation. Using **Python**, the system processes the input data, evaluates crop suitability, and generates intelligent recommendations to support farming decisions.

Key Functions

•Collect soil and environmental parameters (pH, temperature, rainfall, humidity, nutrients).

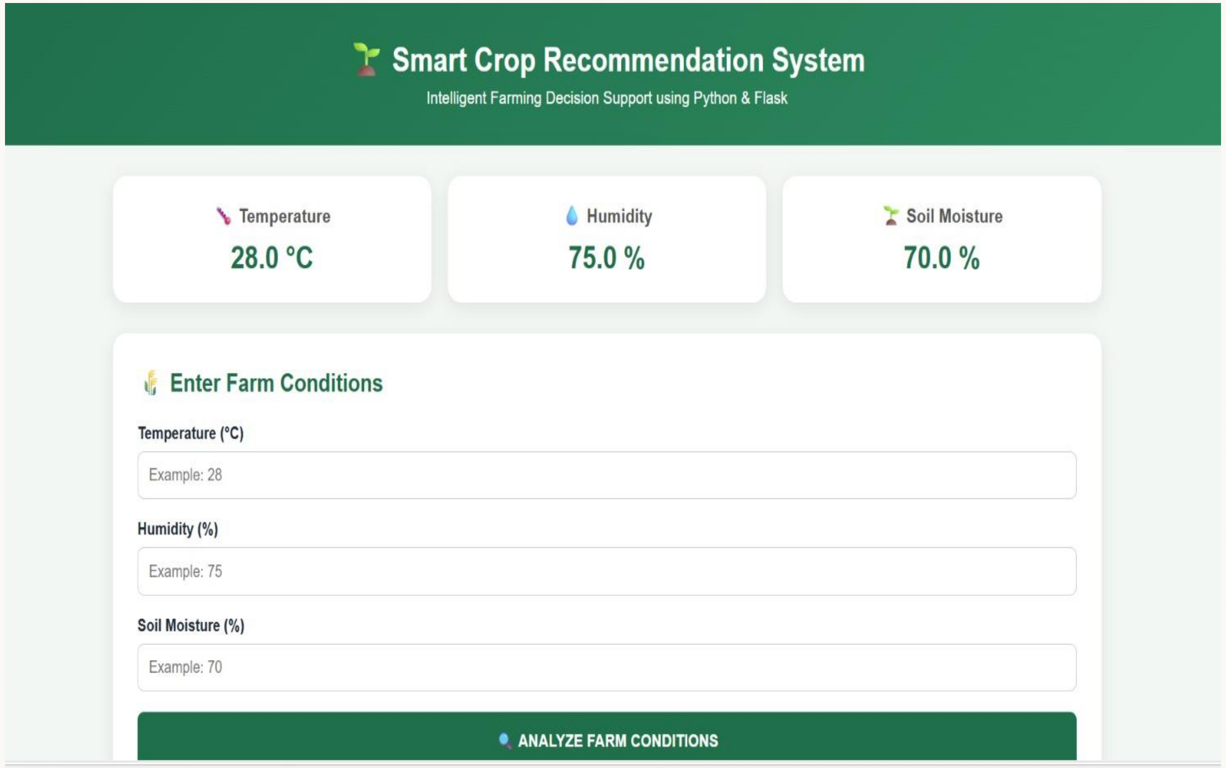
•Process and validate the input data using **Python**.

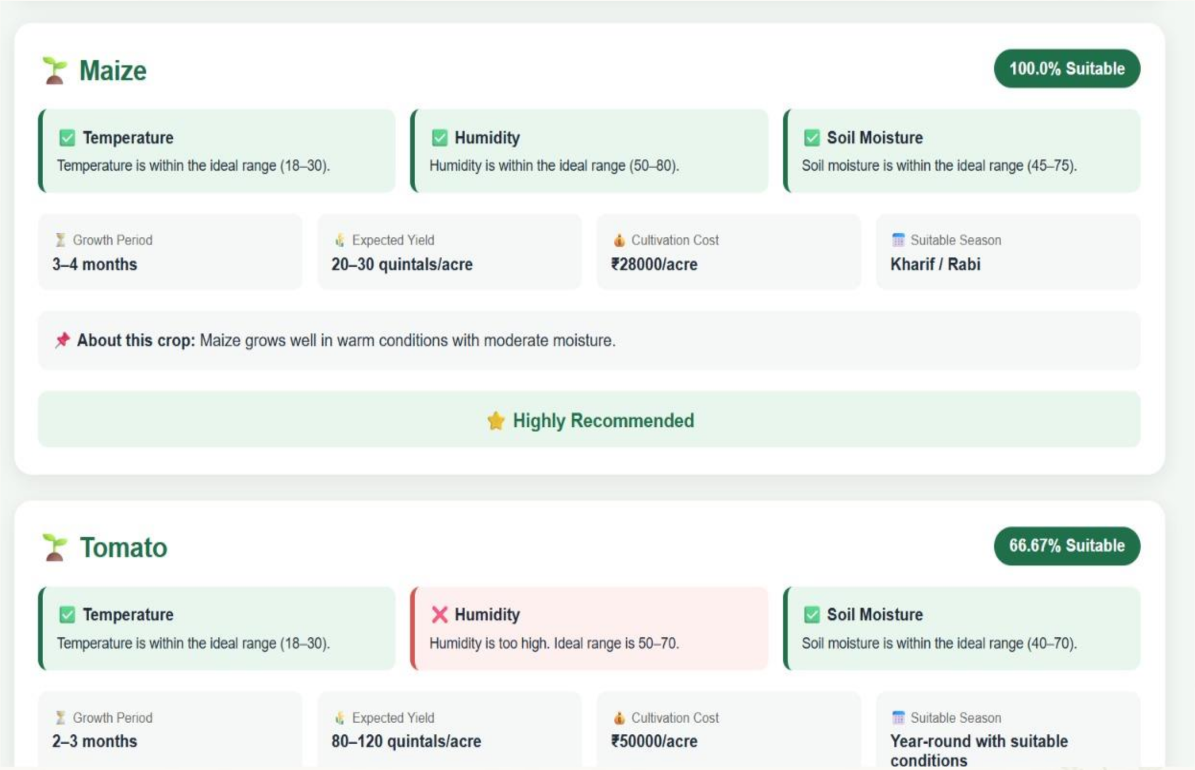
•Calculate plant suitability using decision-making algorithms.

•Rank and recommend the most suitable crops.

•Generate explainable recommendations for better decision support.

Module 1: Output





Module 2: Agricultural Knowledge Repository and Data Processing Layer

Definition

This module serves as a centralized repository for agricultural information. Using **Python**, it stores, organizes, and processes crop-related data such as soil requirements, fertilizers, irrigation methods, and farming practices for efficient retrieval and analysis.

Key Functions

- Store and organize agricultural knowledge.
- Manage crop, soil, and fertilizer information.
- Process and retrieve agricultural data efficiently.
- Support Module 1 with structured and reliable data.
- Enable easy integration of new crops and farming information.

Results & Discussion

- The framework successfully analyzes **soil and environmental conditions** to identify suitable crops.
- It provides **ranked and explainable crop recommendations** to support better farming decisions.
- The agricultural knowledge repository efficiently manages **crop, soil, fertilizer, irrigation, and farming data**.
- The **modular Python architecture** makes the system flexible, scalable, and easy to extend with new crops and information.
- The framework provides a strong foundation for **future Machine Learning and AI integration**, enabling more intelligent farming support.

Conclusion

- The proposed **Modular Python Framework** provides intelligent plant recommendations using soil and environmental data.
- It improves **crop selection and farming decision-making** through efficient data processing.
- The modular design makes the system **scalable, flexible, and easy to extend**.
- Future integration of **Machine Learning and AI** can further improve prediction accuracy and decision support.
- Overall, the framework offers a **practical and reliable approach to smarter and more sustainable farming**.

Reference

1. Sawant, Nitish Rupesh, et al. "An IoT-driven machine learning system for real-time smart crop recommendation and optimization in precision agriculture." *Discover Artificial Intelligence* 6.1 (2026): 194.

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Thank You